1. Given the following instruction sequence for the MIPS processor with the standard 5 stage pipeline,

```
addi    $10, $0, 4
lw      $2, 0($10)
add     $2, $2, $2
sw      $2, 4($10)
```

a. Show the data dependences between the instructions above by drawing arrows between dependent instructions (only show true/data dependencies).

b. **Assuming forwarding** support, in what cycle would the **store instruction write back to memory?** Show the cycle by cycle execution of the instructions as they execute in the pipeline. Also, show any stalls (bubbles) if necessary.

addi:

lw:

add:

sw:

c. For the MIPS pipeline, what is the earliest stage that the control hazard due to the **beq rs, rt, label** instruction can be resolved?

   i. Specify the stage and explain why it can be resolved in that stage but not earlier.

   ii. How many cycles does the pipeline stall due to control hazards?

d. Assuming the following:
   - clock rate of the single cycle machine is 500 MHz
   - clock rate of the pipelined machine is 2GHz
   - processor is running program X from problem 1 where 18% of instructions are branches

   i. Given your answser to part c, ii (number of stalls per branch), assuming no other hazards what is the $CPI_{\text{pipe with branch stalls}}$?

   ii. How much faster does program X run on the pipelined machine with branch stalls than the single cycle machine?